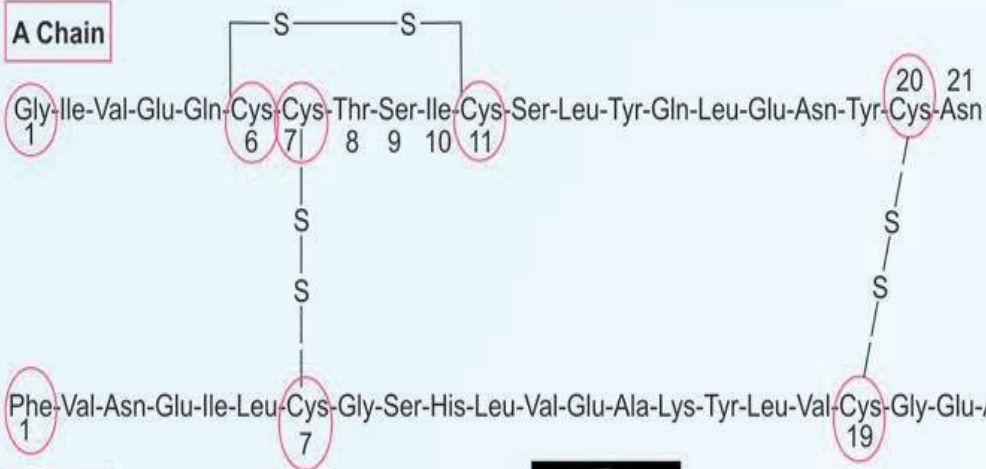


A Chain



Amino acid substitutions

8	9	10 of A chain
Ala-	Ser-	Val (Bovine)
Thr-	Ser-	Ile (Human)
Thr-	Ser-	Ile (Pig)
Ala-	Gly-	Val (Sheep)
Thr-	Gly-	Ile (Horse)

B Chain



Amino acid composition at 30th amino acid in B chain

Thr (Human)
Ala (Bovine)
Ala (Pig)

Frederick Sanger
NP 1958
and 1980



Protein

By: Alie M.

Outline

- Definition and basic structure of protein
- Level of structural organization of protein
- Physical and chemical proprieties of protein
- Classification of protein
- Quantative estimation of protein

Definition

■ The word **protein** is derived from Greek word, "**proteios**" which means primary. As the name shows, the proteins are of paramount importance for biological systems. Out of the total dry body weight, 3/4ths are made up of proteins. Proteins are used for body building; all the major structural and functional aspects of the body are carried out by protein molecules. Abnormality in protein structure will lead to molecular diseases with profound alterations in metabolic functions. Proteins contain Carbon, Hydrogen, Oxygen and Nitrogen as the major components while Sulphur and Phosphorus are minor constituents. **Nitrogen** is characteristic of proteins. On an average, the nitrogen content of ordinary proteins is 16% by weight. All proteins are polymers of amino acids.

■ **Alpha carboxyl group** of one amino acid reacts with **alpha amino group** of another amino acid to form a peptide bond or **CO-NH bridge**. Proteins are made by polymerisation of amino acids through peptide bonds. Two amino acids are combined to form a **dipeptide**; three amino acids form a **tripeptide**; four will make a **tetrapeptide**; a few amino acids together will make an **oligopeptide**; and combination of 10 to 50 amino acids is called as a **polypeptide**. By convention, big polypeptide chains containing more than 50 amino acids are called proteins.

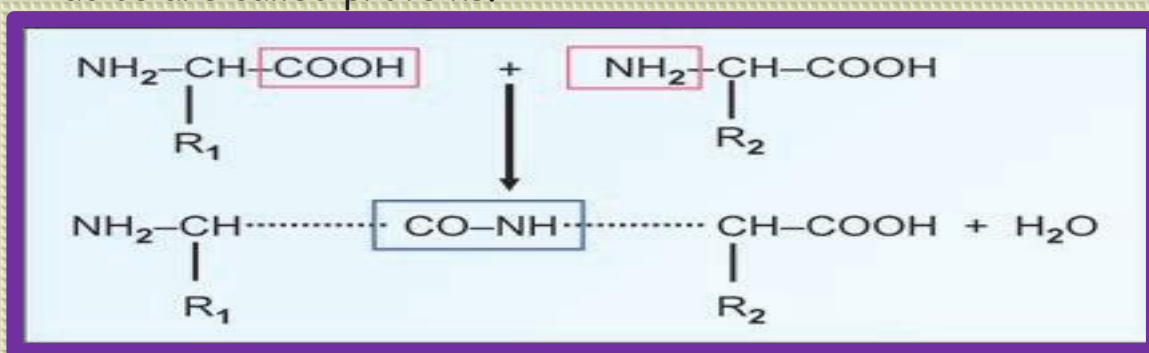


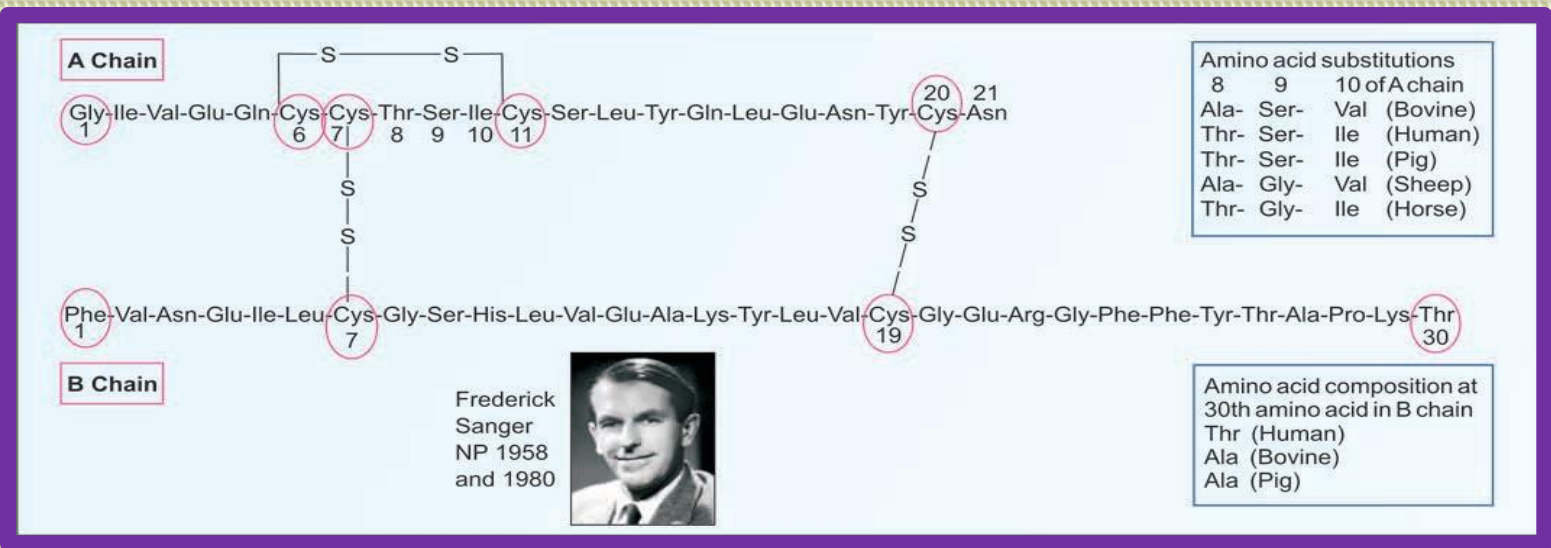
Fig.1: Peptide bond formation

STRUCTURE OF PROTEINS (Organisation of Proteins)

■ Proteins have different levels of structural organisation; primary, secondary, tertiary and quaternary.

I. **Primary structure** ;Denotes the number and sequence of amino acids in the protein. Primary structure of protein means the order of amino acids in the polypeptide chain and the location of disulfide bonds, if any. The higher levels of organisation are decided by the primary structure. Each polypeptide chain has a unique amino acid sequence decided by the genes. The primary structure is maintained by the covalent peptide bonds .

Example: Primary structure of insulin



Frederick Sanger
NP 1958
and 1980



Amino acid substitutions

8	9	10 of A chain
Ala-	Ser-	Val (Bovine)
Thr-	Ser-	Ile (Human)
Thr-	Ser-	Ile (Pig)
Ala-	Gly-	Val (Sheep)
Thr-	Gly-	Ile (Horse)

Amino acid composition at 30th amino acid in B chain

Thr (Human)
Ala (Bovine)
Ala (Pig)

Numbering of Amino Acids in Proteins

In a polypeptide chain, at one end there will be one free alpha amino group. This end is called the **amino terminal (N-terminal)** end and the amino acid contributing the alpha amino group is named as the **first amino acid**. Usually the N-terminal amino acid is written on the left hand side when the sequence of the protein is denoted. Incidentally, the bio synthesis of the protein also starts from the amino terminal end. The other end of the polypeptide chain is the **carboxy terminal end (C-terminal)**, where there is a free alpha carboxyl group which is contributed by the **last amino acid**. All other alpha amino and alpha carboxyl groups are involved in peptide bond formation. Amino acid residues in polypeptides are named by changing the suffix **"-ine"** to **"-yl"**, for example, Glycine to Glycyl. Thus, peptide bonds formed by carboxyl group of glycine with amino group of Alanine, and then carboxyl group of Alanine with amino group of Valine and is called glycyl alanyl-valine and abbreviated as NH₂-Gly-Ala-Val-COOH or Gly-Ala-Val or simply as GAV.

■ **Primary Structure Determines Biological Activity** : A protein with a specific primary structure, when put in solution, will automatically form its natural three dimensional shape. So the higher levels of organization are dependent on the primary structure. Even a single amino acid change (**mutation**) in the linear sequence may have profound biological effects on the function. For example, in HbA (normal hemoglobin) the 6th amino acid in the beta chain is glutamic acid; it is changed to valine in HbS (**sickle cell anemia**).

■ **Secondary structure of protein**: is the steric relationship of amino acids, close to each other. Secondary and tertiary levels of protein structure are preserved by **noncovalent forces** or bonds like **hydrogen bonds**, **electrostatic bonds**, **hydrophobic interactions** and **van der Waals forces**.

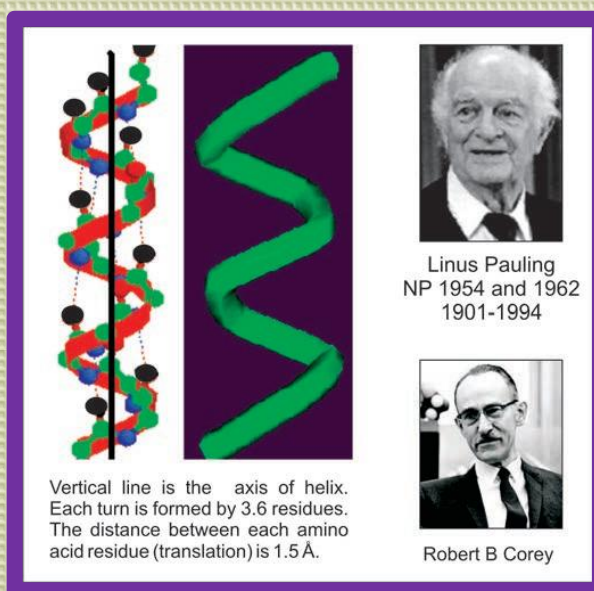
i. A **hydrogen bond** is a weak electrostatic attraction between one electronegative atom like O or N and a hydrogen atom covalently linked to a second electronegative atom. Hydrogen atoms can be donated by -NH (imidazole, indole, peptide); -OH (serine, threonine) and -NH₂ (arginine, lysine). Hydrogen accepting groups are COO⁻ (aspartic, glutamic) C=O (peptide); and S-S (disulphide).

ii. **Electrostatic bonds (ionic bonds)**: Positive charges are donated by epsilon amino group of lysine, guanidinium group of arginine and imidazolium group of histidine. Negative charges are provided by beta and gamma carboxyl groups of aspartic and glutamic acids.

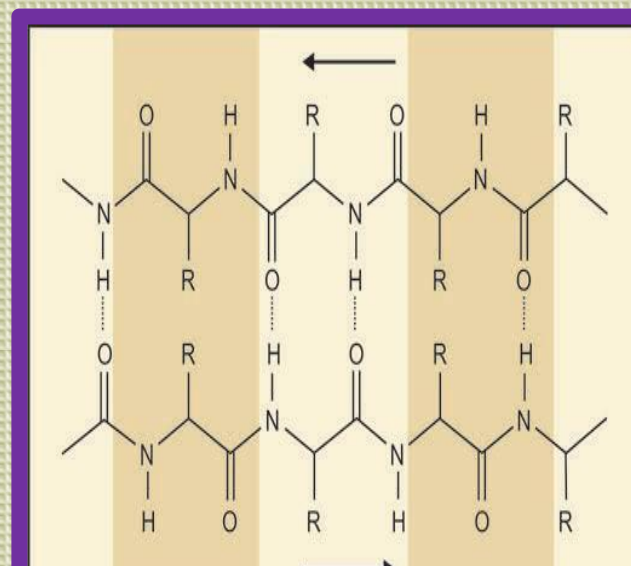
iii. **Hydrophobic bonds** are formed by interactions between nonpolar hydrophobic side chains by eliminating water molecules. This serves to hold lipophilic side chains together.

iv. The **van der Waals forces** are very weak, but collectively contribute maximum towards the stability of protein structure.

■ Secondary structure of protein is manifested as **Alpha helix, Beta plated sheet, collagen helix.**



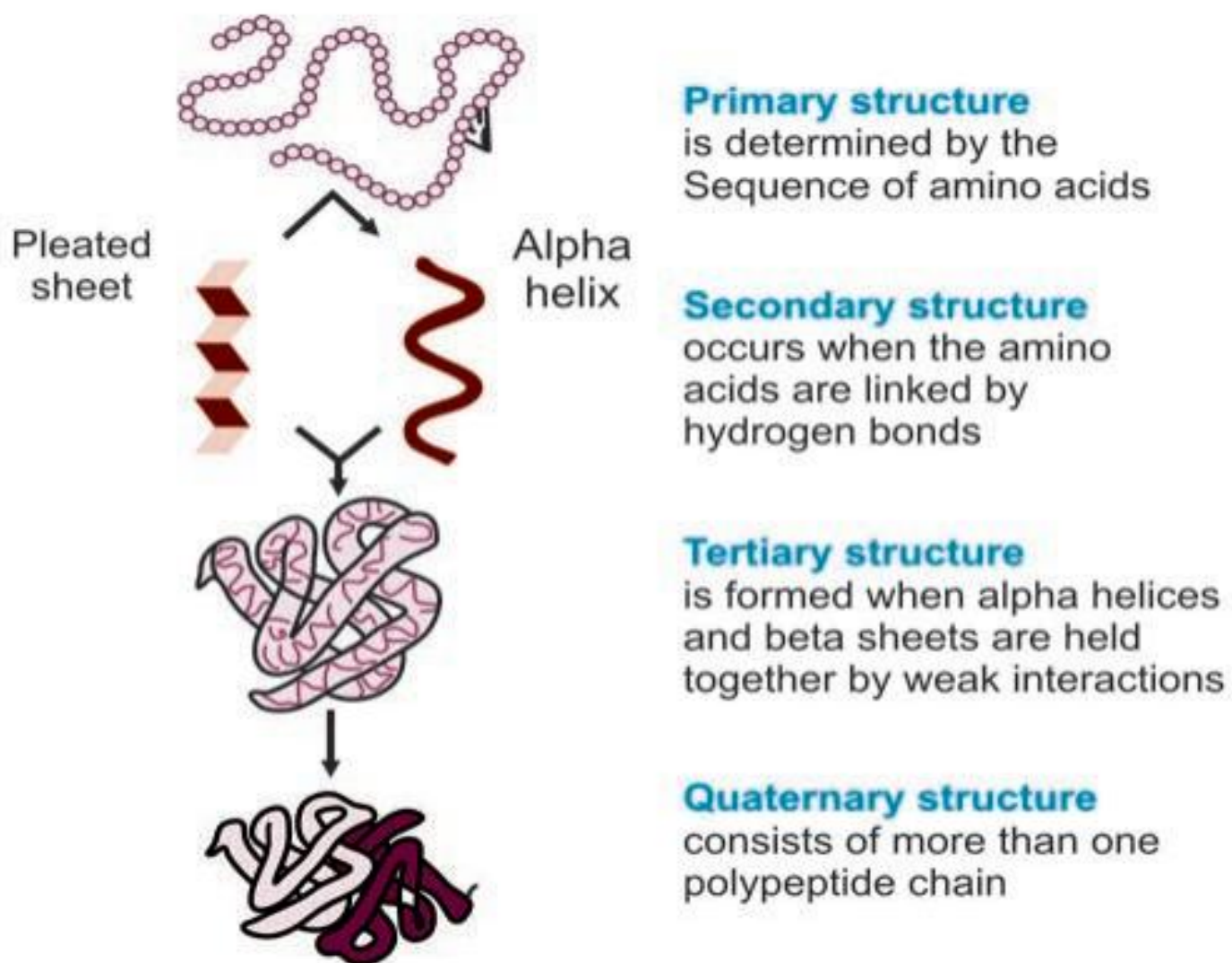
Alpha helix



Beta plated sheet

■ **Tertiary Structure**: denotes the overall arrangement and interrelationship of the various regions, or domains of a single polypeptide chain. It denotes three dimensional structure of the whole protein. The tertiary structure is maintained by **non covalent** interactions such as hydrophobic bonds, electrostatic bonds and van der Waals forces. The tertiary structure acquired by native protein is always thermodynamically most stable.

■ **Quaternary Structure:** results when the proteins consist of two or more polypeptide chains held together by non-covalent forces. The forces that keep the quaternary structure are hydrogen bonds, electrostatic bonds, hydrophobic bonds and van der Waals forces. Depending on the number of polypeptide chain, the protein may be termed as monomer (1 chain), dimer (2 chains), tetramer (4 chains) and so on. Each polypeptide chain is termed as **subunit** or **monomer**. **Homodimer** contains two copies of the same polypeptide chain. **Hetero dimer** contains two different types of polypeptides as a functional unit. For example, 2 alpha-chains and 2 beta-chains form the **Hemoglobin** molecule. Similarly, 2 heavy chains and 2 light chains form one molecule of **immunoglobulin G**. Creatine kinase (CK) is a dimer. Lactate dehydrogenase (LDH) is a tetramer.



Level of organization of protein

PHYSICAL PROPERTIES OF PROTEINS

1. Protein solutions exhibit colloidal properties and therefore scatter light and exert **osmotic pressure**. Osmotic pressure exerted by plasma proteins is clinically important (Chapter 28).
2. **Molecular weights** of some of the proteins are: Insulin (5,700); Hemoglobin (68,000); Albumin (69,000); Immunoglobulins (1,50,000); Rabbit Papilloma Virus Protein (4,70,00,000).
3. **Shape** of the proteins also vary. Thus, Insulin is globular, Albumin is oval in shape, while Fibrinogen molecule is elongated. Bigger and elongated molecules will increase the viscosity of the solution.
4. **Iso-electric pH** Since proteins are made of amino acids, the pI of all the constituent amino acids will influence the pI of the protein.

PRECIPITATION REACTIONS OF PROTEINS

Purification of enzymes and other proteins usually start with precipitating them from solution. The stability of proteins in solution will depend mainly on the charge and hydration. Polar groups of the proteins ($-NH_2$, $COOH$, OH groups) tend to attract water molecules around them to produce a shell of hydration. Any factor which neutralises the charge or removes water of hydration will therefore cause precipitation of proteins. The following procedures are used for protein precipitation:

1. **Salting Out** When a neutral salt such as ammonium sulphate or sodium sulphate is added to protein solution, the shell of hydration is removed and the protein is precipitated. This is called salting out. As a general rule, higher the molecular weight of a protein, the salt required for precipitation is lesser. Thus globulins are precipitated with half saturation of ammonium sulphate; but albumin will need full saturation with ammonium sulphate for complete precipitation.
2. **ISO-electric Precipitation** Proteins are least soluble at their iso-electric pH. Some proteins are precipitated immediately when adjusted to their iso-electric pH. The best example is **Casein** which forms a flocculent precipitate at pH 4.6 and redissolves in highly acidic or alkaline solutions. When milk is curdled, the casein forms the white curd, because lactic acid produced by the fermentation process lowers the pH to the iso-electric point of casein.
3. **Precipitation by Organic Solvents** When an organic solvent is added to the protein solution, water molecules available for proteins are reduced, and precipitation occurs. Organic solvents reduce the dielectric constant of the medium which also favors protein precipitation. Hence, **alcohol** is a powerful protein precipitating agent. This may explain the disinfectant effect of alcohol.

4. **Precipitation by Heavy Metal Ions** In alkaline medium, proteins have net negative charge, or are anions. To such a solution, if salts of heavy metals are added, positively charged metal ions can complex with protein molecules and metal proteinates are precipitated. Salts of **Copper, Zinc, Lead, Cadmium and Mercury** are toxic, because they tend to precipitate normal proteins of the gastro-intestinal wall. Based on this principle, **raw egg** is sometimes used as an **antidote** for mercury poisoning.

5. **Precipitation by Alkaloidal Reagents** : Tungstic acid, Phosphotungstic acid, **Trichloro acetic acid**, Picric acid, Sulphosalicylic acid and **Tannic acid** are powerful protein precipitating agents. These acids lower the pH of medium, when proteins carry net positive charges. These protein cations are complexed with negatively charged ions to form protein-tungstate, protein-picrate, etc. and thick flocculent precipitate is formed. In clinical laboratory phospho-tungstic or trichloro acetic acid are usually used for precipitating proteins. **Tanning** in leather processing is based on the protein precipitating effect of tannic acid. Under certain conditions, proteins undergo **denaturation**, which is a mild form of precipitation reaction. **Heat coagulation** is an irreversible precipitation process.

CLASSIFICATION OF PROTEINS

A. Classification based on functions

1. Catalytic proteins, e.g. enzymes
2. Structural proteins, e.g. collagen, elastin
3. Contractile proteins, e.g. myosin, actin.
4. Transport proteins, e.g. hemoglobin, myoglobin, albumin, transferrin
5. Regulatory proteins or hormones, e.g. ACTH, insulin, growth hormone
6. Genetic proteins, e.g. histones
7. Protective proteins, e.g. immunoglobulins, interferons, clotting factors

B. Classification based on Composition and Solubility

B-1. Simple Proteins: Albumins, Globulins, Protamines, Prolamines, Scleroproteins etc.

B-2. Conjugated Proteins: They are combinations of protein with a non-protein part, called Prosthetic group. Example: lycoproteins, Lipoproteins, Phosphoproteins, Chromoproteins, Nucleoproteins, Metalloproteins

B-3. Derived Proteins: They are degradation products of native proteins. Progressive hydrolysis of protein results in smaller and smaller chains: Protein → peptones → peptides → amino acids.

C. Classification Based on the Shape

C-1. Globular Proteins They are spherical or oval in shape. They are easily soluble. Examples are albumins, globulins and Protamines.

C-2. Fibrous Proteins The molecules are elongated or needle shaped. Their solubility is minimum. They resist digestion. Collagen, elastin and keratins are examples.

D. Classification Based on Nutritional Value

D-1. Nutritionally Rich Proteins They are also called as **complete proteins or first class proteins**. They contain all the essential amino acids in the required proportion. On supplying these proteins in the diet, children will grow satisfactorily. A good example is **casein** of milk.

D-2. Incomplete Proteins They **lack one essential amino acid**. They cannot promote body growth in children; but may be able to sustain the body weight in adults. Proteins from **pulses are deficient in methionine**, while proteins of **cereals lack in lysine**. If both of them are combined in the diet, adequate growth may be obtained.

D-3. Poor Proteins They **lack in many essential amino acids** and a diet based on these proteins will not even sustain the original body weight. Zein from corn lacks tryptophan and lysine.

Biologically Important Peptides

When 10 or less number of amino acids are joined together, it is called an oligopeptide. Some of them are biologically active.

i. **Thyrotropin releasing hormone (TRH)** is a tripeptide with the sequence of Glu-His-Pro; but the Glu is modified to form pyroglutamic acid.

ii. **Glutathione** is a tripeptide. It is gamma glutamyl cysteinyl glycine (See Fig.15.19). It is involved in erythrocyte membrane integrity and is important in keeping enzymes in active state.

iii. **Oxytocin and Vasopressin (ADH)** are nanopeptides; with 9 amino acids. They are secreted by posterior pituitary.

iv. **Angiotensin I** has 10 amino acids and angiotensin II has 8 amino acids. They cause hypertension.

v. **Gramicidin S**, an antibiotic produced by *Bacillus brevis*, contains 10 amino acids. It is circular and contains D-phenyl alanine (usual proteins contain only L-amino acids). Poly peptide hormones (more than 10 amino acids).

QUANTITATIVE ESTIMATION of protein

■ Kjeldahl's Procedure, Biuret Method, Lowry's Method, Spectrophotometric Estimation, Radial Immuno Diffusion (Mancini's technique), Nephelometry, Turbidimetry, RIA and ELISA Tests.

Thank you!!!



